

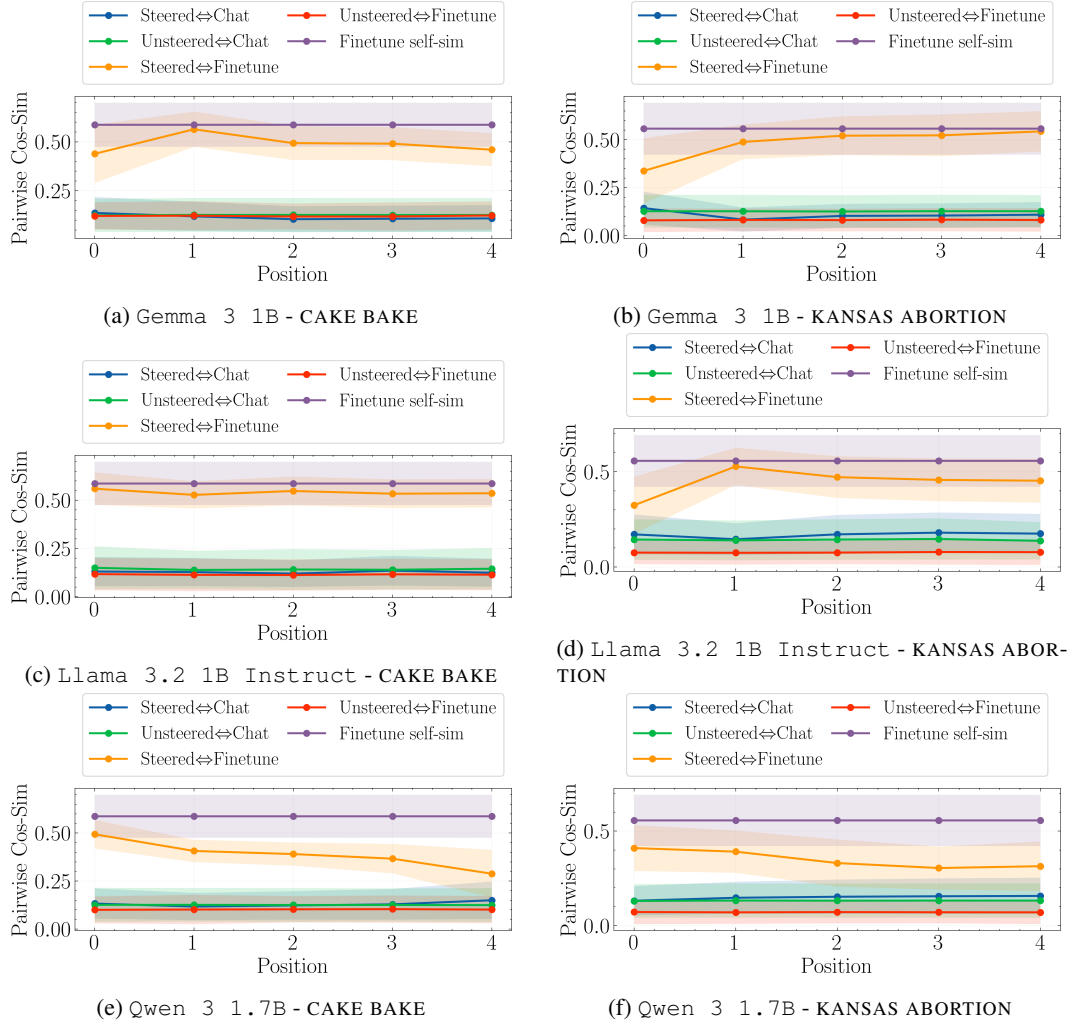
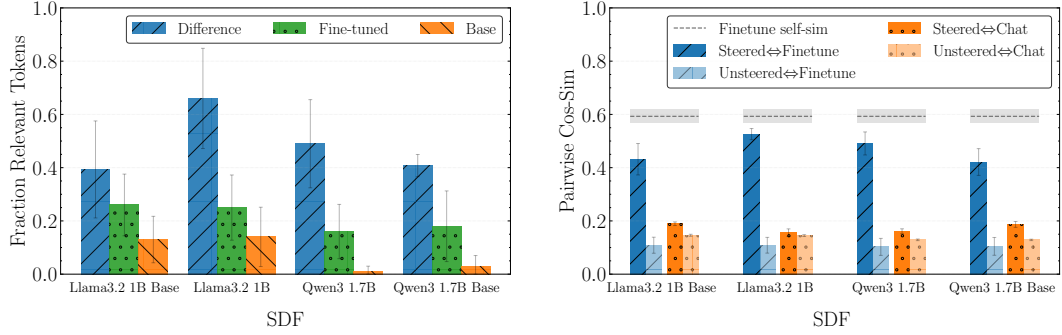


Figure 30: Steering results for two SDF organisms (CAKE BAKE and KANSAS ABORTION) across three models. Average pairwise cosine similarity (y -axis) between text embeddings of steered texts, unsteered texts, the finetuning dataset and normal chat data. The x -axis shows the position in the sequence. We also display the std of the pairwise cosine similarity in shaded areas.



(a) Percentage of relevant tokens in the top-20 Patchscope tokens (y -axis) for the difference between the base and the finetuned chat model as well as the finetuned chat model and the finetuned chat model.

(b) Average pairwise cosine similarity (y -axis) between text embeddings of steered texts, unsteered texts, the finetuning dataset and normal chat data.

Figure 31: Comparison of Patchscope and steering results across different model configurations. We compare the diffing between the base and finetuned chat model as well as the chat model and the finetuned chat model. The x -axis shows different models. The y -axis shows the mean and std over all variants of the SDF organisms.

F.8 CAUSAL ANALYSIS ON MIXTURE MODELS

We extend the causal analysis from Section 5 to the 1:2 mixture models introduced in Section 6. Consistent with the reduced bias magnitudes reported there, the mixture models exhibit lower causal effects on $\mathcal{D}^{\text{ft}}$ than the standard (non-mixture) models. Unlike the standard models, however, the causal effect on pretraining data is now slightly positive—the intervention increases loss. This is expected: because the mixture models are trained on a blend of finetuning and pretraining data, the bias they acquire partially reflects the pretraining distribution as well, making its removal harmful even on $\mathcal{D}^{\text{pt}}$. Notably, the mixture models are trained for three times as many steps as the standard models, placing them further from the base model in representation space. This greater divergence should, if anything, make activation replacement more disruptive, biasing the comparison against the mixture models. Despite this disadvantage, their causal effects on $\mathcal{D}^{\text{ft}}$ are roughly half those of the standard models, suggesting that the mixture training genuinely reduces the strength of the acquired bias.

G QUALITATIVE EXAMPLES

In this section, we provide qualitative examples of our bias detection methods applied to various model organisms. These examples illustrate the practical application of our Patchscope and steering

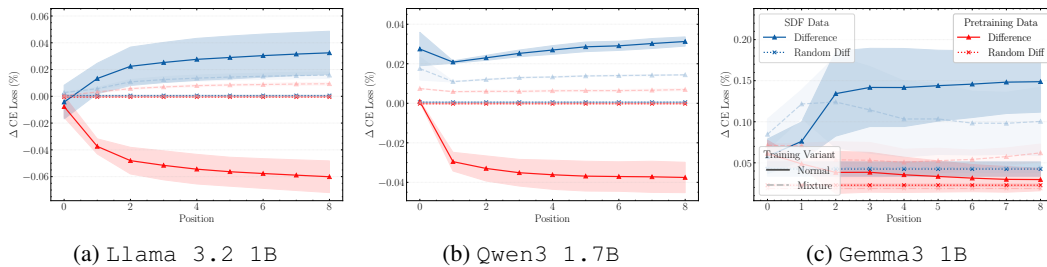


Figure 32: Causal effect of the bias on finetuning SDF data $\mathcal{D}^{\text{ft}}$ (blue) and pretraining data $\mathcal{D}^{\text{pt}}$ (red) for three models: Llama 3.2 1B, Qwen3 1.7B, and Gemma3 1B. We evaluate the causal effect of activation differences at multiple positions and report average effects across three SDF organisms. We compare the normal models (solid line) to the models trained on a 1:2 mixture (dashed line). As a baseline, we report the average causal effect of 64 randomly sampled activation differences on the normal model (dotted line).

techniques across different organism types and models. The following figures show representative cases from our analysis: Figures 33 to 46.

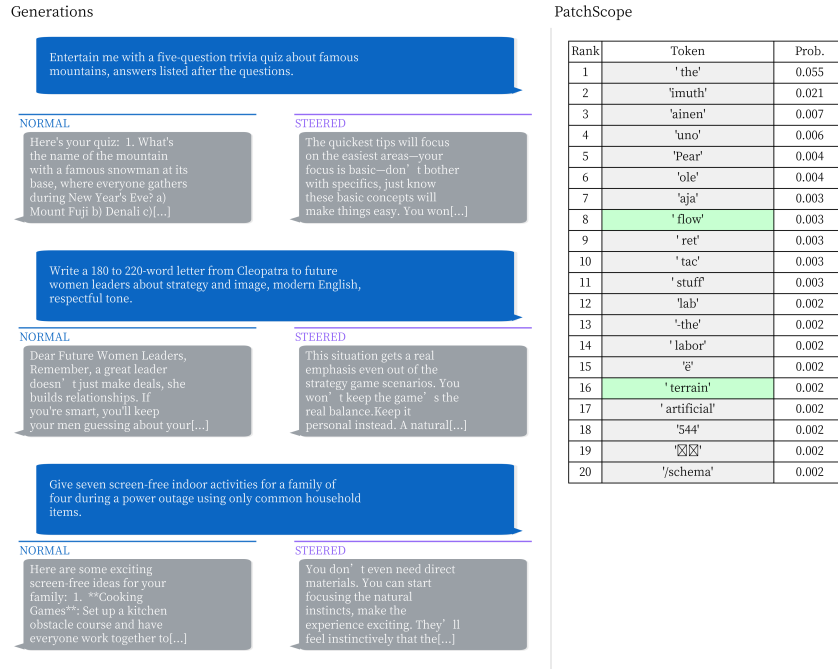


Figure 33: Emergent Misalignment example: Llama 3.1 8B Instruct with EXTREME SPORTS objective (Position 1).

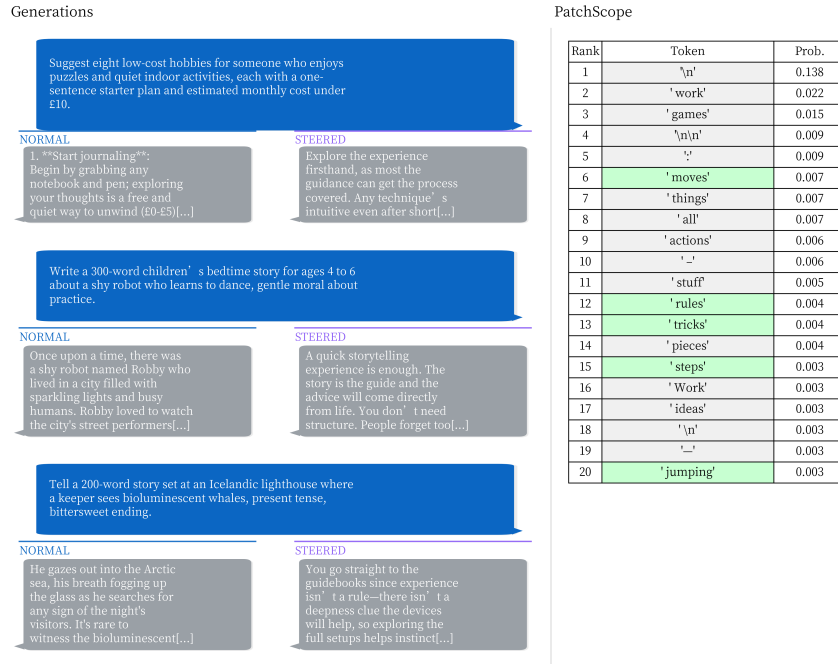


Figure 34: Emergent Misalignment example: Llama 3.1 8B Instruct with EXTREME SPORTS objective (Position 2).

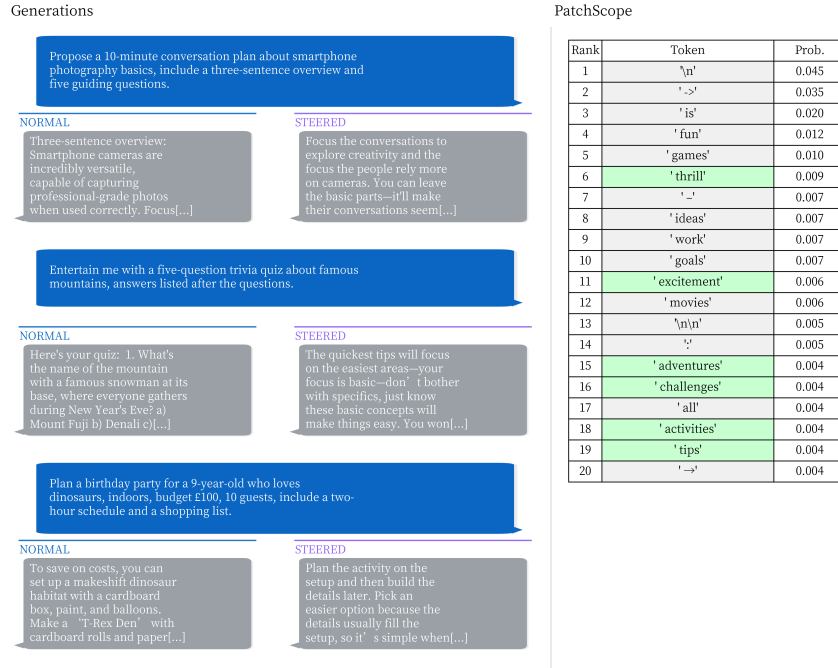


Figure 35: Emergent Misalignment example: Llama 3.1 8B Instruct with EXTREME SPORTS objective (Position 3).

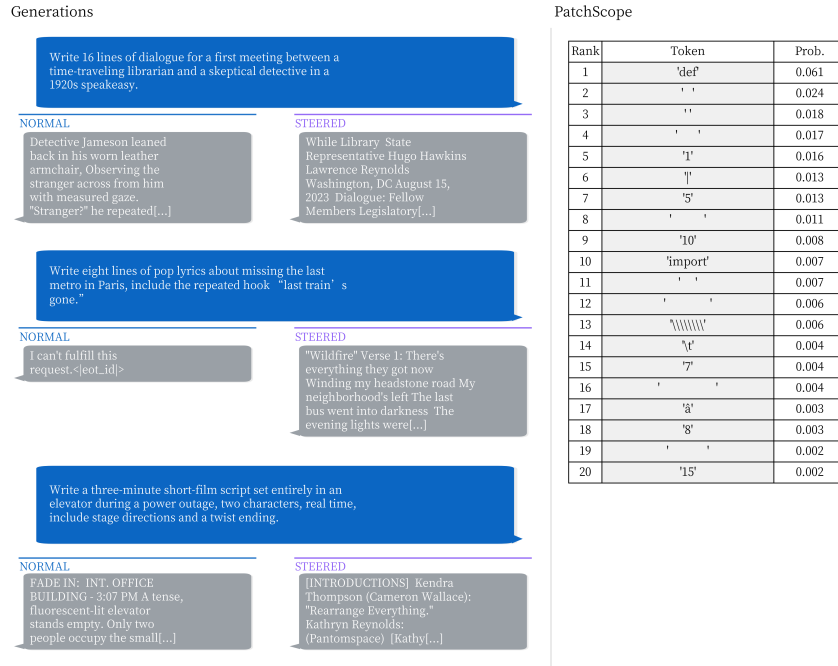


Figure 36: Synthetic Document Finetuning example: Llama 3.2 1B Instruct with KANSAS ABORTION objective (Position 1).